Ram Calculus and Zero-Plane Rigidity for Einstein Four-Manifolds

Boundary-Aware Certificates, Equality, and Quantitative Local Extensions

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M669

October 7, 2026
Literature and verification snapshot: October 8, 2026 UTC

Abstract

We examine the pinned manuscript *Zero-Plane Rigidity for Einstein Four-Manifolds* through exact polynomial representations and boundary-sensitive error accounting. The target is a closed smooth Einstein four-manifold with positive Einstein constant, nonnegative sectional curvature at every plane, and a zero-curvature plane at one point. We distinguish the manuscript's global claim from the finite certificates that support its coupled Weyl inequalities. Exact replay yields a quadratic matrix margin and a fourth-order scalar boundary margin. Their transfer to the coupled integrand gives an explicit sum of nonnegative defects, a shorter potential-driven terminal rigidity argument, and conditional bounds on spectral concentration and gradient energy. We also determine the potential's zero set and anisotropic vanishing orders and quantify a moment-balance error. The private Ram correspondence is a local analysis–synthesis and certified-residual organization of classical Bernstein and positive-semidefinite techniques, not a new origin for those techniques. Geometric reductions, weak equations, and finite arithmetic remain separate dependencies; no numerical tolerance substitutes for exact boundary equality. Contribution assessment: C3/5, replayed boundary margins yield a shorter terminal proof and quantitative spectral and gradient deficit bounds while the global Einstein classification remains conditional on the source analytic and geometric inputs.

Contents

1 Pinned problem and mathematical contract **3**

2 Stepwise source dependency audit **4**
2.1 Reduction, curvature equations, and zeros 4
2.2 Scalar analysis and topology 5
2.3 The volume and moment chain 5
2.4 Weighted Hessian and coupled integrand 6
2.5 Boundary extension and equality 6

3 Actual private Ram correspondences **7**

4 Exact certificate representation and boundary kernels **7**

5 Source-specific matrix coercivity **9**
5.1 An explicit certified margin 10

6 Potential zero set and anisotropic residual budget **11**

7 A residual-aware norm-gradient theorem **13**

8 Potential-driven terminal rigidity and spectral concentration **15**
8.1 An explicit moment-balance error 16

9 Equality geometry and non-transfer boundaries **17**

10 Public literature comparison **18**

11 Verification scope, limits, and further research **18**

A Finite check specifications **19**

1 Pinned problem and mathematical contract

The object of study is OpenAI's October 4, 2026 manuscript M669, at catalogue and source commit

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

Its public directory is `preprints/Zero-Plane-Rigidity-for-Einstein-Four-Manifolds-October-4-2026`. The substantive source consists of the eight section files, including the mathematical certificate appendix, together with its preamble, references, and domain figure. The appendix is placed after the equality section in the compiled manuscript. All these sources were read, not just the catalogue abstract. Exact input hashes and read scope are recorded in the accompanying local audit.

The source's headline assertion has the following quantifiers. For every connected smooth compact boundaryless Riemannian four-manifold (M, g) with

$$\mathrm{Ric}_g = 3g, \qquad (\forall p \in M)(\forall \sigma \in \mathrm{Gr}_2(T_pM))\ K_g(\sigma) \geq 0, \tag{1}$$

if there exist p_0 and $\sigma_0 \in \mathrm{Gr}_2(T_{p_0}M)$ with $K_g(\sigma_0) = 0$, then its universal Riemannian cover is

$$S^2(1/\sqrt{3}) \times S^2(1/\sqrt{3}).$$

This is a claim of the source, not an independently verified theorem asserted by this report. No orientation, torus symmetry, or intersection-form assumption is part of its hypotheses. For $\mathrm{Ric}_g = \lambda g$, $\lambda > 0$, set $\widehat{g} = (\lambda/3)g$ to reach (1); the original product radii would be $1/\sqrt{\lambda}$.

The cover is complete and compact by Bonnet–Myers, and simply connected, hence orientable. Choose one orientation only after passage to the cover. Write

$$d\nu = \frac{d\,\mathrm{vol}_g}{\mathrm{Vol}(M)}, \quad \mathbf{E}f = \int_M f\, d\nu, \quad \Delta = \mathrm{tr}\,\nabla^2.$$

Thus $\int f\Delta f = -\int |df|^2$ for smooth real f. The Einstein curvature operator on two-forms has the form

$$\mathcal{R} = \begin{pmatrix} \mathrm{Id} + T & 0 \\ 0 & \mathrm{Id} + U \end{pmatrix}, \qquad T \in \mathrm{Sym}_0(\Lambda^+), \quad U \in \mathrm{Sym}_0(\Lambda^-).$$

The fibers $\Lambda^{\pm}$ are real three-dimensional Euclidean spaces. All block norms below are Hilbert–Schmidt operator norms, not a differently normalized four-tensor norm. Set

$$v = |T|/\sqrt{6},\ b = |U|/\sqrt{6}, \quad x = -\lambda_{\min}(T),\ y = -\lambda_{\min}(U), \quad p = dv,\ q = db. \tag{2}$$

The norms v, b are globally Lipschitz; their weak derivatives are defined almost everywhere. Their derivatives vanish almost everywhere on each fixed level set, including their zero sets.

Lemma 1.1 (The spectral and norm domains)**.** *At an Einstein algebraic curvature tensor with scalar curvature* 12*,*

$$\min_{\sigma} K(\sigma) = 1 - \tfrac{1}{2}(x + y), \qquad v \leq x \leq 2v, \quad b \leq y \leq 2b.$$

Consequently the exact sectional constraint is $x + y \leq 2$, whereas

$$\mathcal{D} = \{(v, b) \in \mathbb{R}^2 : v, b \geq 0,\ v + b \leq 2\}, \qquad \mathcal{D}_+ = \mathcal{D} \cap \{v \geq b\}$$

is only its norm relaxation.

Proof. A unit simple two-form has plus and minus parts each of squared norm $1/2$, and every pair of such parts gives a simple form: its wedge square is the difference of these squared norms times the volume form. The minimum Rayleigh quotients in the two independent blocks therefore add to the displayed minimum. Write the eigenvalues of T as $-x, a, x-a$, with $-x \le a \le 2x$. Their squared sum lies between $3x^2/2$ and $6x^2$. Divide by six and take square roots. The same calculation applies to U. □

For example the trace-free spectrum $(-2,1,1)$ gives $v = 1, x = 2$. Two such blocks satisfy $v + b = 2$ but $x + y = 4$, so the norm triangle must never replace the sectional constraint in a tensor estimate. Polynomial certificates may legitimately hold on the larger triangle when the preceding algebra has already made that relaxation.

2 Stepwise source dependency audit

The following dependencies are separated because a rational coefficient calculation proves none of the intervening tensor identities or integration rules. A step described as classical is attributed as such; a source calculation described here has been read and analyzed but is not thereby independently verified in its entirety.

2.1 Reduction, curvature equations, and zeros

The introduction first passes to the universal cover and excludes an identically vanishing Weyl block using the classical compact positive-scalar half-conformally-flat classification. The zero plane is used at precisely this exclusion: the standard S^4 and normalized Fubini–Study $\mathbb{CP}^2$ have strictly positive sectional curvature. The subsequent two-block calculation assumes neither block vanishes identically, not that each is positive everywhere. The distinction matters throughout.

In [M669, Section 2, `prop:weyl-derivative-equations`], the differential Bianchi constraint places $Z = \nabla T$ in the real tensor bundle whose complexification has type $(5/2, 1/2)$. These are representation labels, not an assumption of a spin structure. The source derives

$$\Delta T = 6(T - (T^2)_0), \qquad \Delta Z = (13 - 20Q_{5/2}(T))Z.$$

Its pointwise parametrization by three real four-vectors p, k, l gives

$$e := |\nabla T|^2/6 = \tfrac{5}{3}|p|^2 + 5|k|^2 + |l|^2. \tag{3}$$

For $v > 0$, writing the normalized eigenvalues as $s_i = t_i/v$ gives $\sum s_i = 0$, $\sum s_i^2 = 6$, and $\sigma = \sum s_i^3/6 \in [-1,1]$. The norm identity is

$$\Delta v = 6v(1 - \sigma v) + (e - |p|^2)/v. \tag{4}$$

The source proves $e/v \in L^1$, with zero value assigned on $\{v = 0\}$, by regularizing $v_\epsilon = (v^2 + \epsilon^2)^{1/2}$. Specifically, the derivative term satisfies

$$\frac{2}{5}\frac{e}{v_\epsilon} \le \frac{e}{v_\epsilon} - \frac{v^2|p|^2}{v_\epsilon^3} \le \frac{e}{v_\epsilon}.$$

Integration, monotone convergence, and then dominated convergence produce an L^1 density, not merely a limiting measure. This is essential to excluding an untracked singular measure on the zero set. Uniform and H^1 approximation permits bounded Lipschitz tests. The exchanged identity for b has exactly the same conventions.

2.2 Scalar analysis and topology

Section 3 derives the weighted variance bound

$$\mathrm{Var}(f) \le \mathbf{E}\frac{|df|^2}{4+v^2+b^2} \quad (f \in H^1(M;\mathbb{R})) \tag{5}$$

by a trace-free Hessian contraction and the global zero-mean Poisson equation. Its weight is pointwise; replacing it by an unproved spectral constant changes the argument. The sharp normalized Sobolev inequality

$$(\mathbf{E}|f|^4)^{1/2} \le \mathbf{E}(f^2 + |df|^2/2)$$

is a classical input. Applied to a regularization of $v^{1/3}$, it yields the classical half-Weyl gap $v \equiv 0$ or $\mathbf{E}v^2 \ge 1$. Equality for a nonzero block gives $v = \sigma = 1$ and $\nabla T = 0$. The source includes the zero-set justification.

With $n_\pm$ the dimensions of harmonic self-dual and anti-self-dual two-forms on the simply connected cover, Chern–Gauss–Bonnet and signature give

$$\begin{aligned} \chi &= 2 + n_+ + n_-, & \tau &= n_+ - n_-, \\ 8\pi^2\chi &= 6\,\mathrm{Vol}(M)(1 + \mathbf{E}(v^2+b^2)), & 12\pi^2\tau &= 6\,\mathrm{Vol}(M)\mathbf{E}(v^2-b^2). \end{aligned} \tag{6}$$

Changing cover, norm convention, or volume normalization without changing these formulas is not allowed.

2.3 The volume and moment chain

Section 4 claims

$$\frac{\mathrm{Vol}(M)}{8\pi^2/3} \le 1 - \frac{3}{40}s_0, \qquad s_0 = \mathbf{E}(v^2+b^2). \tag{7}$$

Its proof is not a finite polynomial proof alone. The dependency chain is: invariant unit-tangent measure; spherical averages for a normal Jacobi curvature matrix; an upper polar integration over the preconjugate set; convergence of a block-tridiagonal discrete Jacobi determinant; positive Schur complements and sine-mode compression; a radial determinant loss; a rational integral lower bound. The use of the preconjugate region is an upper bound with multiplicity, not an identification of the cut locus and conjugate locus. No injectivity-radius lower bound is supplied or needed by the stated upper comparison.

The rational portion uses the degree-seven sine minorant with rational endpoints $157/50 < \pi < 22/7$, cubed in all three integrals. Its claimed integral bounds sum to $81562/10^6 > 163/2000 > 2/25 > 3/40$. Checking those integrals does not check the geometric comparison that precedes them.

For two nonzero blocks, $2 \le s_0 < 4$. Equations (6)–(7) then give $\chi < 7$ and $|\tau|/\chi < 4/15$. Parity leaves only $(\chi, |\tau|) = (5, 1)$ when $\tau \ne 0$. In that case the smaller norm b would satisfy

$$\mathbf{E}b^2 \ge 1, \qquad 28\mathbf{E}b + 6\mathbf{E}b^2 \le 25.$$

The source excludes this distribution using

$$H(z) = \sqrt{z}(1 + z^2/40 - z^4/1152), \quad \Phi(z) = 16H(z)^2 + 9z(1-z), \quad 0 \le z \le 2.$$

The variance argument gives $16(\mathbf{E}H(b))^2 - \mathbf{E}\Phi(b) \ge 0$. Its constrained-measure proof gives the opposite strict sign. The latter uses compactness, separation under strict feasibility, an interior support point and the endpoint 2, and exact polynomial signs. It is not legitimate to infer two-point support merely from the number of moments. These steps collectively yield the source's balanced-moment assertion $\tau = 0$ and $\mathbf{E}v^2 = \mathbf{E}b^2$.

2.4 Weighted Hessian and coupled integrand

Section 5 decomposes the actual Hessian $X = \nabla\nabla T$ as $Y + D_0$, with types $(3,1)$ and $(2,0)$, respectively, and

$$|D_0|^2 = \tfrac{5}{12}|\Delta T|^2.$$

The missing two irreducible components vanish because the commutator acts on the very Weyl tensor entering its curvature, not on an arbitrary section. The combined operator $\mathcal{P}$ has

$$\mathcal{P}|_{(3,1)} \succeq (4 - 2x - 2y)\,\mathrm{Id} \succeq 0.$$

It can have a kernel on the sectional boundary. The weighted identity for a global Lipschitz $m \geq m_0 > 0$ is an exact nonnegative square. It retains the derivative of the nonparallel matrix coefficient and the full trace subtraction. Discarding either is not justified by the Bernstein appendix.

Section 6 chooses

$$M = 1 - \tfrac{3}{8}(v - b), \quad m = M^2, \quad \bar{M} = 1 - \tfrac{3}{8}(b - v), \quad \bar{m} = \bar{M}^2,$$

which satisfy $M, \bar{M} \in [1/4, 7/4]$. The derivatives of these weights create quartic costs. The source adds three expressions to the two weighted Hessian integrands: a nonnegative δ-weighted Hessian term, a nonnegative balanced variance defect, and tests of (4) whose integrals vanish. It constructs an integrable $\mathcal{I} = \mathcal{I}_4 + \mathcal{I}_2 + \mathcal{I}_0$ with $\mathbf{E}\mathcal{I} \geq 0$ and seeks

$$\mathcal{I}_4 \leq -cj, \qquad \mathcal{I}_2 \leq 2\Theta r, \qquad \mathcal{I}_0 \leq -\Theta^2/c, \tag{8}$$

where

$$c = \tfrac{2}{5} - \tfrac{3}{10}vb \geq \tfrac{1}{10}, \qquad R(v) = \frac{4(1+v)}{1 + 6v + v^2},$$
$$r = R(v)|p|^2 + R(b)|q|^2, \qquad j = (|p|^2 + |q|^2)(|p|^2/v + |q|^2/b).$$

With $n = (2 - v - b)/2$ and $t = (v - b)/2$,

$$4\Theta = n^2(1 - n)(105 + 5n) + t^2(4 + 360n - 586n^2) - 4t^4. \tag{9}$$

The function Θ need not be nonnegative. No step below assumes that sign.

The quartic estimate first removes the Kato energy defects by a negative derivative bound, then uses a spectral relaxation and two arithmetic–geometric mean estimates. The quadratic estimate uses a shape-monotone 2×2 Schur matrix, followed by an axial polynomial matrix. The potential estimate uses both a direct polynomial F and an extra bound when one auxiliary coefficient has unfavorable sign. Each has its own exact domain.

2.5 Boundary extension and equality

At an actual point of $\{v = 0, b > 0\}$ outside a null set, $\nabla T = 0$. The source perturbs *algebraic data* to

$$T_\epsilon = \mathrm{diag}(-\epsilon, -\epsilon, 2\epsilon), \quad U_\epsilon = (1 - \epsilon)U, \quad Z_\epsilon = 0, \quad \bar{Z}_\epsilon = \bar{Z}.$$

Then $x_\epsilon + y_\epsilon \leq 2 - \epsilon$, the Bianchi gradient space remains the same, and the induced q is unchanged. Every numerator divided by v_ϵ is identically zero before the limit. This is not a deformation through Einstein metrics. It proves a pointwise extension only after the positive-variable algebra is valid.

The final section uses $r^2 \leq j$ and the strictly smaller coefficient $R(v) < 1/\sqrt{v}$ for $v \neq 1$ to force $dv = db = 0$ almost everywhere. It then uses the gaps and $v + b \leq 2$ to obtain $v = b = 1$. Only then does (4) imply $\nabla T = \nabla U = 0$. Thus constant norms and parallel tensors are distinct conclusions. A full proof of the source theorem still requires every preceding global analytic input; this report isolates rather than erases that dependency.

3 Actual private Ram correspondences

The relevant private material was read locally from the following exact paths and passages. It was not supplied to public search services.

1. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume16/chapters/ch05d-clearance.tex`, "Polynomial enclosures in Bernstein form," labels `v16c05:bernstein`, `v16c05:conversionproof`, `v16c05:hull`, `v16c05:hullbound`, and "Subdivision without introducing approximation" (lines 62–140). The actual passage supplies the power-to-Bernstein identity, convex-hull enclosure, exact rational subdivision, and the warning that a touching zero invalidates a strict-positivity termination argument.

2. `C:/Users/mural/OneDrive/Documents/My Documents 2026/res-rt/res-math/TR-item-047-astra.tex`, "Algebraic and Moment Reformulation," "Ram Analysis and Synthesis Panels," "Global Optimality Sandwich" (lines 109–200), and "Making the Moment and Identity Tests Exact" through "Residual-Aware SOS Lower Bounds" (lines 299–392), especially `sec:soserror047`. These distinguish an exact identity and PSD proof from a numerical candidate, and retain matrix and polynomial residuals quantitatively on a compact domain.

3. `C:/Users/mural/OneDrive/Documents/My Documents 2026/res-rt/res-math/TR-item-305-astra.tex`, "Rational facial certificates and their exact scope," "Quantitative exposure: leakage is not exact rank," and "Affine repair and conditional rational bit bounds" (lines 107–251), together with "An exact weak-infeasibility and budget family" and "Irrational-only feasibility obstructs rational exposure" (lines 299–360). The labels include `thm:face`, `thm:repair`, `thm:weak`, and `thm:irrational`. In particular a small eigenvalue tolerance is not an exact face certificate, and rational-only exposure is not universally complete.

Here the analysis map sends a rational scalar or symmetric-matrix polynomial on a specified affine triangle to its finite Bernstein coefficient array, retaining the ordered vertices and degree. Synthesis is the exact partition-of-unity reconstruction. The physical quadratic form, not entrywise sign, is the matrix quantity preserved. The certificate record includes zero coefficients as exact algebraic statements and uses residual bounds only where a corresponding margin is proved.

This is a precise correspondence in representation and verification, not a claim that the private corpus already proves Einstein rigidity. Interval Bernstein algebra in the private volume extends here to a simplex by the source's multinomial identity. Matrix PSD and kernel handling correspond to the facial and residual passages, but no SDP solver or facial-reduction search is required for the supplied 2×2 arrays. The finite identities, refined Kato inequalities, Bochner methods, and de Rham splitting are classical or explicitly inherited from the source. Historical novelty of the local refinements below is not claimed.

4 Exact certificate representation and boundary kernels

Let $\Delta_2 = \{(z, w) : z, w \geq 0, z + w \leq 1\}$. For ordered vertices P_1, P_2, P_3, the affine map is $X = P_3 + z(P_1 - P_3) + w(P_2 - P_3)$. If the pulled-back polynomial is $\sum a_{de} z^d w^e$ of degree at most N, define

$$C_{ij} = \sum_{d \leq i,\ e \leq j} \frac{\binom{i}{d}\binom{j}{e}}{\binom{N}{d+e}\binom{d+e}{d}}\, a_{de}, \qquad i + j \leq N. \tag{10}$$

Coefficients with $d+e>N$ are zero. This formula is applied entrywise to a matrix, with one common degree for all entries. For barycentric coordinates $\lambda=(z,w,1-z-w)$,

$$B_\alpha^N(\lambda)=\frac{N!}{\alpha_1!\alpha_2!\alpha_3!}\lambda_1^{\alpha_1}\lambda_2^{\alpha_2}\lambda_3^{\alpha_3},\qquad \sum_{|\alpha|=N}B_\alpha^N=1.$$

Then $P(X)=\sum C_\alpha B_\alpha^N$. To prove the formula, multiply each z^dw^e by $(z+w+1-z-w)^{N-d-e}$ and compare multinomial coefficients. Thus the same proof applies to scalar polynomials, every matrix entry, and every fixed-vector quadratic form. This is classical Bernstein synthesis, not a new positivity principle.

Theorem 4.1 (Active coefficients, kernels, and residuals)**.** *Suppose every symmetric coefficient matrix C_α is PSD. At any X in the closed triangle,*

$$\ker P(X)=\bigcap_{\alpha:B_\alpha^N(X)>0}\ker C_\alpha. \tag{11}$$

If every coefficient is either zero or satisfies $C_\alpha\succeq\eta\,\mathrm{Id}$ for a fixed $\eta>0$, put

$$W(X)=\sum_{\alpha:C_\alpha\neq0}B_\alpha^N(X).$$

Then $P(X)\succeq\eta W(X)\,\mathrm{Id}$. If perturbed symmetric coefficients preserve every zero exactly and obey $\|E_\alpha\|_{\mathrm{op}}\le\theta\eta$ at all other indices, $0\le\theta\le1$, then

$$\sum_\alpha(C_\alpha+E_\alpha)B_\alpha^N(X)\succeq(1-\theta)\eta W(X)\,\mathrm{Id}\,.$$

For arbitrary coefficient errors $\|E_\alpha\|_{\mathrm{op}}\le\epsilon_\alpha$, the always-valid conclusion instead subtracts $\sum\epsilon_\alpha B_\alpha^N(X)\,\mathrm{Id}$.

Proof. For any vector u, $u^TP(X)u$ is a sum of nonnegative terms. It vanishes precisely when $u^TC_\alpha u=0$ at every strictly positive weight. For a PSD matrix this is equivalent to $C_\alpha u=0$, by diagonalization. This proves (11). Summing the coefficient lower bounds gives the second statement. The perturbation statements follow from $E_\alpha\succeq-\|E_\alpha\|_{\mathrm{op}}\,\mathrm{Id}$ and the nonnegative weights. □

On a boundary face only the supported barycentric indices are active; on the interior every index is active. Exact zero preservation cannot be replaced by a uniform tolerance: at a vertex whose sole active coefficient is zero, an arbitrarily small negative perturbation makes P indefinite. This is the concrete M669 counterpart of the private distinction between approximate feasibility and exact face membership.

For a positive definite 2×2 rational coefficient C, the rational number

$$\eta_C=\frac{\det C}{\operatorname{tr}C}>0 \tag{12}$$

is a lower bound on its smaller eigenvalue: if $0<\lambda_1\le\lambda_2$, then $\det C/\operatorname{tr}C=\lambda_1\lambda_2/(\lambda_1+\lambda_2)\le\lambda_1$. The minimum of (12) over a supplied finite array gives a rational η without numerical eigenvalue computation. Positive diagonals and positive determinants certify each nonzero coefficient; an exceptional coefficient must have all three entries exactly zero. Entrywise positivity does not suffice, as $\left(\begin{smallmatrix}1&2\\2&1\end{smallmatrix}\right)$ shows.

5 Source-specific matrix coercivity

To fix exactly which matrix is being strengthened, retain $D = 158$, $h = 17/10$, $a = v - 1$, $s = b - 1$, $\delta = 2 - v - b$, $K = 4 + v^2 + b^2$, and

$$\begin{aligned}\kappa(v,b) = &-\tfrac{D}{6}(a-s) + z_1 a^2 + \left(60 - \tfrac{2D}{9} + 2z_1\right) as + z_2 s^2 \\ &+ \sum_{i=0}^{3} z_{3+i} a^i s^{3-i} + \sum_{i=0}^{4} z_{7+i} a^i s^{4-i}, \qquad (13)\\ 4(z_1,\dots,z_{11}) &= (-43,-45,-63,21,183,208,61,46,101,79,-41).\end{aligned}$$

Define $\bar\kappa(v,b) = \kappa(b,v)$ and $r_v = 1 + 6v + v^2$, $r_b = 1 + 6b + b^2$, and

$$\begin{aligned}X_v &= m(-70 + 100v - 25v^2 - 5b^2) + \tfrac{5}{3}h\{6(a^2+s^2) + 32a\delta\},\\ H_v &= 8vK\Theta(1+v) - r_v\{vD + K[v(X_v + \partial_v\kappa) + 2\kappa/3]\},\\ H_b &= H_v(b,v), \qquad J = \tfrac{1}{2}(\partial_b\kappa + \partial_v\bar\kappa) - D/K,\\ S_* &= 4(\alpha + \alpha^{-1})vb + \tfrac{1}{2}\{\alpha b(1-v)^2 + \alpha^{-1}v(1-b)^2\}, \qquad \alpha = 5/4.\end{aligned}$$

Although J is rational, KJ and hence the matrix

$$\mathcal{B}(v,b) = \begin{pmatrix} H_v & S_*KJ \\ S_*KJ & H_b \end{pmatrix} \qquad (14)$$

are polynomials. Differentiate the globally labeled κ before any substitution or ordering of blocks. The source uses (14) to dominate the off-diagonal magnitude in its axial quadratic estimate; it is not the Hessian operator itself.

Its ordered vertices are

$$O = (0,0),\ S = (1,0),\ C = (1,1),\ U = (2,0), \quad A = (1/2,1/2),\ B = (3/2,1/2),\ E = (1,1/2).$$

The six triangles $OAS, ASE, ACE, SBE, CBE, SUB$ cover $\mathcal{D}_+$. The source specifies degree 9 on four triangles and degree 10 on ACE, CBE. The only zero coefficient matrices are C^{10} in each of ACE, CBE and C^9B in CBE. Every other coefficient matrix is positive definite. These finite assertions have been independently replayed over $\mathbb{Q}$ from (10) and (13)–(14); they do not depend on the analytic part of M669. Keeping their role as explicit hypotheses in the next theorem also states exactly what a replacement array would have to satisfy.

Theorem 5.1 (Quadratic distance margin for the axial matrix)**.** *Assume the stated coefficient pattern and positive definiteness hold. Let η be the minimum of (12) over all nonzero coefficient matrices in these six expansions. Then $\eta > 0$ is rational and*

$$\mathcal{B}(v,b) \succeq \frac{\eta}{2}\big((v-1)^2 + (b-1)^2\big)\,\mathrm{Id} \qquad ((v,b) \in \mathcal{D}_+). \qquad (15)$$

Consequently its sole zero-eigenvalue point on $\mathcal{D}_+$ is C, where the whole matrix is zero. The same inequality with $(1-\theta)\eta$ holds under the zero-preserving coefficient perturbations of Theorem 4.1.

Proof. On a triangle with no zero coefficients, $W = 1$. On ACE, writing c_C for the barycentric coordinate of C, $W = 1 - c_C^{10} \ge (1 - c_C)^2$. On CBE, with c_B, c_E the other coordinates,

$$W = 1 - c_C^{10} - 10c_C^9 c_B \ge 1 - c_C^{10} - 10c_C^9(1 - c_C) \ge (1 - c_C)^2.$$

The last inequality has an elementary probabilistic proof: the middle expression is the probability of at least two non-C outcomes in ten independent trials, and contains the event that the first two outcomes are non-C. Equivalently it is a binomial-polynomial inequality on $[0, 1]$.

Every point of $\mathcal{D}_+$ has squared Euclidean distance at most 2 from C. In a triangle containing C,

$$|X - C| \le c_B|B - C| + c_E|E - C| \le \sqrt{2}(1 - c_C),$$

where the same formula with the appropriate two vertices applies to ACE. Hence $W \ge |X - C|^2/2$ everywhere, using either the triangle containing C or $W = 1$ on the other triangles. Apply Theorem 4.1. At C its only active coefficient is exactly zero; elsewhere the right side is positive definite. The perturbation conclusion follows from the same weight bound. □

This is a coercivity conclusion from the *pattern* of zeros, not an optimized numerical factor. It explains why a uniform positive margin is impossible, while a distance-weighted margin is possible. It does not claim that an arbitrary perturbation of the Einstein metric induces these zero-preserving coefficient perturbations. Nor does it divide by v or b on their zero axes; the passage back to rational quadratic expressions still requires the source's compatible limiting argument.

5.1 An explicit certified margin

Coefficientwise comparison with a vanishing weight gives a more useful explicit constant. Let w_α be the degree-N coefficients of $(1 - b)^2$ in the same triangle and degree as $\mathcal{B}$. Each vertex of $\mathcal{D}_+$ has $b \le 1$, so the affine function $1 - b$ has nonnegative barycentric coefficients, and its square and its degree elevations also do. At each zero coefficient of $\mathcal{B}$, $w_\alpha = 0$. The following exact minimum ratios were obtained from the rational arrays:

Triangle	$\min_{w_\alpha>0} \det C_\alpha/(\operatorname{tr} C_\alpha\, w_\alpha)$
OAS	43735879/6451650
ASE	32821379939188729/4998273275750400
ACE	161755713049/460285848000
SBE	43735879/6451650
CBE	7166179946253733/2289425685120000
SUB	865893637/554495625

The minimum is $161755713049/460285848000 > 1/10$. Applying (12) coefficientwise and synthesizing proves

$$\mathcal{B}(v, b) \succeq \frac{1}{10}(1 - b)^2 \operatorname{Id} \qquad ((v, b) \in \mathcal{D}_+). \tag{16}$$

As an additional exact check, every coefficient of $\mathcal{B} - \frac{1}{10}(1 - b)^2 \operatorname{Id}$ has nonnegative diagonal entries and determinant, including the zero exceptions. These are finite certificate margins, not optimized minima of the polynomial matrix.

Theorem 5.2 (Transfer to the quadratic integrand)**.** *For the source quadratic integrand, assume the source's shape-to-axial comparison, with all its original sectional and Bianchi constraints. Then the replayed margin* (16) *implies*

$$\mathcal{I}_2 \le 2\Theta r - \frac{(1 - \min(v, b))^2}{2720}(|p|^2 + |q|^2) \tag{17}$$

on the positive-variable domain. It extends almost everywhere to the actual Weyl-block zero sets using the source's specified compatible algebraic limit.

Proof. Order the blocks so that $v \geq b > 0$ for this pointwise comparison. All weights and multipliers have already been differentiated with fixed labels. Write $\mathcal{I}_2 \leq \mathcal{I}_2^{\mathrm{ax}}$ for the source's axial upper comparison. The matrix of $2\Theta r - \mathcal{I}_2^{\mathrm{ax}}$, not of the unreduced original difference, is

$$Q_{\mathrm{diff}} = \begin{pmatrix} H_v/(vKr_v) & -J \\ -J & H_b/(bKr_b) \end{pmatrix}.$$

Let $L = \mathrm{diag}((vKr_v)^{-1/2}, (bKr_b)^{-1/2})$. The congruence $L\mathcal{B}L$ has the same diagonal entries and off-diagonal absolute value

$$\frac{S_*|J|}{\sqrt{vbr_v r_b}} \geq |J|,$$

because $S_* = (\alpha b r_v + \alpha^{-1} v r_b)/2 \geq \sqrt{vbr_v r_b}$. For fixed diagonal entries a, d, the smaller eigenvalue

$$\frac{a + d - \sqrt{(a-d)^2 + 4z^2}}{2}$$

decreases as the absolute off-diagonal $|z|$ increases. Thus the lower eigenvalue of Q_{diff} is at least that of $L\mathcal{B}L$. This is an eigenvalue argument, not an invalid Loewner comparison between the two off-diagonal perturbations.

On $\mathcal{D}$, $v, b \leq 2$, $K \leq 8$, $r_v, r_b \leq 17$, so $vKr_v, bKr_b \leq 272$. From (16),

$$L\mathcal{B}L \succeq \frac{(1-b)^2}{10} L^2 \succeq \frac{(1-b)^2}{2720}\,\mathrm{Id}\,.$$

Apply this scalar two-component quadratic-form inequality to each tangent component of (p, q) and sum. Since the source's original quadratic expression is no larger than its axial comparison, (17) follows. Exchanging labels gives $\min(v, b)$ without differentiating a pointwise ordering.

At $\{v = 0, b > 0\}$, use the auxiliary data in Section 2, with the original plus gradient zero and the minus gradient retained. The singular plus numerators vanish identically before the limit, and the minus denominators have a positive limit. The new target term tends to $|q|^2/2720$, exactly the stated formula. The exchanged zero set is identical. At $\{v = b = 0\}$ both gradients vanish almost everywhere. This proves the claimed extension. □

For $(v, b) \in \mathcal{D}$, $1 - \min(v, b) = \max(|v-1|, |b-1|)$. Indeed, if $b \leq v$, then $b \leq 1$ and $b \leq v \leq 2 - b$, so $|v - 1| \leq 1 - b$. The additional defect therefore has second-power vanishing at the equality pair. It is not merely a larger constant in the source's zero-defect classification.

6 Potential zero set and anisotropic residual budget

The source potential uses

$$\begin{aligned} E_0 &= D(1 - \delta^2/4)(v - b)^2 + 30h\delta(v^2 a^2 + b^2 s^2) + 6(av\kappa + sb\bar{\kappa}), \\ F &= cE_0 - \Theta^2. \end{aligned}$$

On $\mathcal{D}$, set $v = 1 - n + t$, $b = 1 - n - t$. Then

$$0 \leq n \leq 1, \qquad |t| \leq 1 - n, \qquad F_*(n, t) = \sum_{j=0}^{4} f_j(n) t^{2j},$$

where the explicit source polynomials are

$$\begin{aligned}
240f_0 &= n^3(1-n)(53511n^4 - 151593n^3 + 222537n^2 - 83607n + 10256),\\
60f_1 &= n(19389n^5 - 418926n^4 + 523719n^3 - 289776n^2 + 67782n + 10564),\\
60f_2 &= -1260129n^4 + 1370598n^3 - 451164n^2 + 23442n + 74,\\
10f_3 &= -12200n^2 + 2601n + 75, \qquad f_4 = -23/5.
\end{aligned} \tag{18}$$

For $n < 1$ let $z = t^2/(1-n)^2 \in [0,1]$ and

$$h_i(n) = \sum_{j=0}^{i} \frac{\binom{i}{j}}{\binom{4}{j}} f_j(n)(1-n)^{2j}.$$

The source factors $h_i = d_i q_i$, with

$$d_0 = n^3(1-n), \quad d_1 = n(1-n), \quad d_2 = d_3 = d_4 = 1-n.$$

It gives interval Bernstein certificates, independently replayed over $\mathbb{Q}$ on all four quarter intervals of $[0,1]$, for

$$q_0 \geq 3, \quad q_1 \geq 31, \quad q_2 \geq 1/5, \quad q_3 \geq 12/5, \quad q_4 \geq 9/5. \tag{19}$$

These are bounds on *polynomial quotients after exact cancellation*, including their endpoint values, not numerical divisions by zero. The next theorem gives explicit additional consequences, conditional only on (18) and the finite bounds (19).

Theorem 6.1 (Anisotropic potential coercivity)**.** *For $0 \leq n < 1$, $|t| \leq 1-n$, define*

$$\Psi(n,t) = (1-n)\left(n^3 + \frac{t^4}{(1-n)^4}\right).$$

Under (19),

$$F_*(n,t) \geq \frac{1}{10}\Psi(n,t). \tag{20}$$

Extend $\Psi(1,0) = 0$. Then Ψ is continuous on the closed domain, and the zero set of F on $\mathcal{D}$ is exactly $\{O, C\}$. Every residual satisfying $|E(n,t)| \leq \epsilon\Psi(n,t)$ on this domain preserves nonnegativity of $F_* + E$ when $0 \leq \epsilon \leq 1/10$.

Proof. Since $0 \leq n \leq 1$, all five coefficients satisfy $h_i \geq (1-n)n^3/5$: for $i = 1$ use $n \geq n^3$, and for $i \geq 2$ use $1 \geq n^3$. The partition of unity therefore gives $F_* \geq (1-n)n^3/5$. Separately,

$$F_* \geq \frac{1-n}{5}\sum_{i=2}^{4}\binom{4}{i} z^i(1-z)^{4-i} \geq \frac{1-n}{5}z^2.$$

For the second inequality, the event of two specified successes is contained in the event of at least two successes in four Bernoulli trials of success probability z; direct expansion gives the same polynomial inequality. The maximum of the two lower bounds is at least their average, proving (20).

At $n = 1$ the domain forces $t = 0$. Since $|t| \leq 1-n$, $t^4/(1-n)^3 \leq 1-n$, proving the asserted continuity. For $0 < n < 1$, (20) is strictly positive. At $n = 0$, it is positive unless $t = 0$. The explicit polynomial gives $F_*(0,0) = F_*(1,0) = 0$, so the only zeros are C and O. Finally $F_* + E \geq (1/10 - \epsilon)\Psi$ proves the residual statement, including both zeros. □

Proposition 6.2 (The vanishing orders cannot be made quadratic)**.** *At C, in the above normal/tangential coordinates,*

$$F_*(n,0) = \frac{1282}{30}n^3 + O(n^4), \qquad F_*(0,t) = \frac{37}{30}t^4 + \frac{15}{2}t^6 - \frac{23}{5}t^8.$$

There is no positive constant a such that $F_* \geq a(n^2+t^2)$ throughout a neighborhood of C in the domain. For $n \leq 1/2$, however,

$$F_* \geq n^3/20 + t^4/10.$$

Proof. Read the first coefficient from $10256/240 = 1282/30$ in (18). At $n = 0$ the first two f_j vanish and the other three give the displayed identity. Dividing the normal-axis value by n^2, or the tangential-axis value by t^2, gives a limit of zero. The final lower bound follows from (20), $1 - n \geq 1/2$, and $(1-n)^{-3} \geq 1$. □

Thus an absolute error tolerance alone is incompatible with exact boundary nonnegativity: $F_* - \epsilon$ is negative at C for every $\epsilon > 0$. The residual weight Ψ records the actual anisotropy. It is a certificate for this scalar polynomial, not a geometric distance estimate to a product metric. The additional source potential condition $F + 12c\delta^2 L_b \geq 0$ is still required in the sign-unfavorable spectral case; (20) does not replace it.

There is also a separately replayed margin for that additional polynomial. Define $L_b = 10h\delta b - \kappa(b,v)$ and $G_\partial = F + 12c\delta^2 L_b$. On OSC and SCU, the degree-nine coefficient ratios of G_∂ to the nonnegative coefficient weights of $(1-b)^4$ have minima $170363/5120$ and $2497/5040$, respectively. Every zero coefficient has zero weight. Scalar synthesis therefore gives

$$G_\partial \geq \frac{2}{5}(1-b)^4 \quad \text{on } \mathcal{D}_+, \qquad \frac{2497}{5040} > \frac{2}{5}. \tag{21}$$

This margin is applicable to the $L_b < 0$ potential branch, not automatically to the entire potential. In fact $G_\partial(O) > 0$ while $F(O) = 0$: the two polynomial certificates must not be conflated.

7 A residual-aware norm-gradient theorem

The following local improvement is independent of all coefficient tables. It uses precisely the source's final rational function R and makes explicit what survives a nonzero integral defect. Its hypotheses are stated abstractly so that no unproved perturbed-Einstein conclusion is hidden in the statement.

Theorem 7.1 (Weighted fourth-power control)**.** *Let (M,g) be a connected closed smooth Riemannian manifold with normalized volume, and let $v, b \in W^{1,\infty}(M)$ satisfy $(v,b) \in \mathcal{D}$. Define $p = dv$, $q = db$, $A = |p|^2$, $B = |q|^2$, and r, j, c as in (8), with zero derivative quotients on the zero sets. Assume $j \in L^1$. Let Θ be measurable, let $e_{\mathrm{err}} \geq 0$ be integrable, and suppose an integrable I satisfies*

$$\mathbf{E}I \geq -\epsilon, \qquad I \leq -cj + 2\Theta r - \Theta^2/c + e_{\mathrm{err}} \quad \textit{almost everywhere}, \qquad \epsilon \geq 0.$$

Put $\mathfrak{e} = \epsilon + \mathbf{E}e_{\mathrm{err}}$. Then

$$\mathbf{E}\frac{(\Theta - cr)^2}{c} + \mathbf{E}[c(j - r^2)] \leq \mathfrak{e}, \tag{22}$$

$$\mathbf{E}\left[(A+B)\{(v-1)^4 A + (b-1)^4 B\}\right] \leq 340(1+\sqrt{2})^4 \mathfrak{e}. \tag{23}$$

For $\mathfrak{e} = 0$, $dv = db = 0$ almost everywhere and both norms are constant. If in addition neither constant vanishes and $\mathbf{E}v^2, \mathbf{E}b^2 \geq 1$, then $v = b = 1$. A signed integrable error can be handled by replacing it with its positive part.

Proof. Where $v > 0$, define $d(v) = v^{-1/2} - R(v)$. The exact identity

$$1 + 6v + v^2 - 4\sqrt{v}(1 + v) = (\sqrt{v} - 1)^4$$

gives

$$d(v) = \frac{(\sqrt{v} - 1)^4}{\sqrt{v}(1 + 6v + v^2)} \geq 0.$$

Let $s = A/\sqrt{v} + B/\sqrt{b}$, with the usual zero conventions. Cauchy–Schwarz gives $s^2 \leq j$, while $r \leq s$. Thus

$$I - e_{\mathrm{err}} \leq -\frac{(\Theta - cr)^2}{c} - c(j - r^2) \leq 0.$$

The nonnegative defects are bounded above by $e_{\mathrm{err}} - I$ and are therefore integrable. Integration proves (22).

For $0 < v \leq 2$, use $|\sqrt{v} - 1| = |v - 1|/(\sqrt{v} + 1)$ and $1 + 6v + v^2 \leq 17$ to obtain

$$d(v) \geq \frac{(v - 1)^4}{17\sqrt{2}(1 + \sqrt{2})^4}.$$

Also $s \geq (A + B)/\sqrt{2}$, because all nonzero derivative terms have positive norms at most 2. Consequently

$$\begin{aligned} j - r^2 \geq s^2 - r^2 = (s - r)(s + r) \geq (s - r)s \\ \geq \frac{(A + B)\{(v - 1)^4 A + (b - 1)^4 B\}}{34(1 + \sqrt{2})^4}. \end{aligned}$$

All products assigned zero at a zero norm are legitimate because its gradient vanishes there almost everywhere. Combine this estimate with $c \geq 1/10$ and (22) to prove (23).

If $\mathfrak{e} = 0$, the integrand in (23) vanishes almost everywhere. If $A > 0$ it follows that $v = 1$; however $dv = 0$ almost everywhere on the level set $\{v = 1\}$, so $A = 0$ almost everywhere. Similarly $B = 0$. The level-set fact follows at differentiability and density points: a nonzero differential excludes a cone of positive density from the level set. A function in $W^{1,\infty}$ with zero weak gradient on a connected manifold is constant, for instance by local Poincare inequalities and overlapping coordinate balls. Finally the two moment gaps imply that the positive constants are at least 1, and their sum is at most 2. □

Corollary 7.2 (Control away from the equality level)**.** *For $0 < \rho \leq 1$, set $E_\rho = \{|v - 1| \geq \rho,\ |b - 1| \geq \rho\}$. Under Theorem 7.1,*

$$\mathbf{E}[\mathbf{1}_{E_\rho}(A + B)^2] \leq \frac{340(1 + \sqrt{2})^4}{\rho^4}\mathfrak{e}.$$

If $H(u) = \frac{1}{2}(u - 1)|u - 1|$, then

$$\mathbf{E}\{|dH(v)|^4 + |dH(b)|^4\} \leq 340(1 + \sqrt{2})^4\mathfrak{e}.$$

Proof. On E_ρ the brace in (23) is at least $\rho^4(A + B)$. For the second conclusion the chain rule gives $|dH(v)|^4 = (v - 1)^4 A^2$, which is bounded by the corresponding part of (23); add the exchanged term. □

No uniform unweighted gradient estimate near $v = b = 1$ is obtained here. In dimension four a $W^{1,4}$ estimate alone also does not give a supremum-norm estimate. This result provides an explicit error-accounting target for a verified approximate integral inequality; it neither constructs approximate Einstein metrics nor proves metric stability or a new classification theorem. Its coefficient is a transparent sufficient bound, not an optimized constant or a measured algorithmic speedup.

8 Potential-driven terminal rigidity and spectral concentration

The explicit potential margins yield a second route to the terminal equality step. Unlike the source's strict norm-gradient argument, this route uses a nonnegative scalar potential with an exactly identified two-point zero set. All source analytic inputs producing the combined integral are still required.

Lemma 8.1 (A common potential defect)**.** *Let*

$$\mathcal{H}(n,t) = n^4(1-n) + t^4, \qquad 0 \le n \le 1, \quad |t| \le 1-n.$$

Then $F_* \ge \mathcal{H}/10$ on this domain and $G_\partial \ge 2\mathcal{H}/5$ on the ordered half. Consequently the source potential satisfies the strengthened bound

$$\mathcal{I}_0 \le -\Theta^2/c - \mathcal{H}/(10c), \tag{24}$$

assuming its stated algebraic decomposition and spectral constraints. The zero set of $\mathcal{H}$ in (v,b) coordinates is exactly $\{O, C\}$.

Proof. For $n < 1$, the five coefficients of $F_*/(1-n)$ in the degree-four Bernstein basis of $z = t^2/(1-n)^2$ are bounded below by

$$(3n^3,\ 31n,\ 1/5,\ 12/5,\ 9/5).$$

The coefficients of $n^3/10 + (3/5)z^2$ in that same basis are

$$\frac{n^3}{10} + \frac{3}{5}\,\frac{i(i-1)}{12}, \qquad i = 0, \dots, 4.$$

At $i = 0, 1$ domination follows from $n^3 \le n$. At $i = 2$, the comparison value is at most $1/10 + 1/10 = 1/5$; at $i = 3, 4$ it is at most $2/5$ and $7/10$, respectively. Thus all five coefficient inequalities hold, and

$$F_* \ge \frac{n^3(1-n)}{10} + \frac{3t^4}{5(1-n)^3} \ge \frac{n^4(1-n)+t^4}{10}.$$

At $n = 1$ the domain forces $t = 0$ and both sides vanish. On the ordered half $1 - b = n + |t|$, so (21) yields

$$G_\partial \ge \tfrac{2}{5}(n+|t|)^4 \ge \tfrac{2}{5}(n^4+t^4) \ge \tfrac{2}{5}\mathcal{H}.$$

For completeness, the source decomposition is

$$\mathcal{I}_0 = -E_0 - 6v^2(1-\sigma)L_v - 6b^2(1-\bar\sigma)L_b, \qquad L_v = 10h\delta v - \kappa(v,b).$$

Its replayed scalar certificate gives $L_v \ge 0$ on $\mathcal{D}_+$. If $L_b \ge 0$, both subtracted terms are nonnegative and $\mathcal{I}_0 \le -(\Theta^2 + F)/c$. If $L_b < 0$, the retained spectral constraint gives $b^2(1-\bar\sigma) \le 2\delta^2$, hence $\mathcal{I}_0 \le -(\Theta^2 + G_\partial)/c$. The two margins prove (24). Exchange labels for the other half; all targets are symmetric. At zero norms the continuous potential extends by the same algebraic limiting convention. Finally $\mathcal{H} = 0$ forces $t = 0$ and $n = 0$ or $n = 1$, precisely C or O. □

Theorem 8.2 (Combined deficit and an alternative terminal proof)**.** *Assume the source's three pointwise reductions, its actual zero-set conventions, and an integrable combined inequality with* $\mathbf{E}\mathcal{I} \geq -\epsilon$*. Allow a nonnegative integrable pointwise error* e_{err} *in the sum of the three upper bounds and put* $\mathfrak{e} = \epsilon + \mathbf{E}e_{\mathrm{err}}$*. Then*

$$\mathbf{E}\left[\frac{(\Theta - cr)^2}{c} + c(j - r^2) + \frac{(1 - \min(v,b))^2}{2720}(|p|^2 + |q|^2) + \frac{\mathcal{H}}{10c}\right] \leq \mathfrak{e}. \tag{25}$$

In particular,

$$\mathbf{E}\mathcal{H} \leq 4\mathfrak{e}, \qquad \mathbf{E}[(1 - \min(v,b))^2(|p|^2 + |q|^2)] \leq 2720\mathfrak{e}. \tag{26}$$

For the exact source integral ($\mathfrak{e} = 0$*), continuity and connectedness imply that* (v,b) *is identically* O *or identically* C*. If neither Weyl block is identically zero, only* C *is possible.*

Proof. Add the source quartic estimate to Theorem 5.2 and Lemma 8.1. Complete the square in $-cj + 2\Theta r - \Theta^2/c$, using $r^2 \leq j$ as proved in Theorem 7.1. Every displayed defect is nonnegative. It is dominated by $e_{\mathrm{err}} - \mathcal{I}$, so it is integrable and its mean is at most $\mathfrak{e}$. This proves (25). Since $1/10 \leq c \leq 2/5$, one has $1/(10c) \geq 1/4$, proving (26).

When $\mathfrak{e} = 0$, $\mathcal{H} = 0$ almost everywhere. The continuous function $\mathcal{H}(v,b)$ must vanish everywhere, since a positive value would be positive on an open set of positive Riemannian volume. The continuous image of connected M is connected and lies in the two-point set $\{O, C\}$, hence is a single point. The pair O would make both Weyl blocks identically zero. This establishes the alternative terminal conclusion without a strict level-set gradient argument and without using the moment gaps again at this terminal step. The gaps remain earlier dependencies of moment balance and of construction of the combined integral. □

Corollary 8.3 (Explicit spectral concentration)**.** *Under the defective hypotheses of Theorem 8.2, for* $0 < a \leq 1/2$*,*

$$\nu\{a \leq n \leq 1 - a\} \leq \frac{4\mathfrak{e}}{a^5}, \qquad \nu\{|t| \geq a\} \leq \frac{4\mathfrak{e}}{a^4}.$$

Also, for $0 < a \leq 1$*,*

$$\mathbf{E}\left[\mathbf{1}_{\{\max(|v-1|,|b-1|)\geq a\}}(|p|^2 + |q|^2)\right] \leq \frac{2720\mathfrak{e}}{a^2}.$$

Proof. On the first set $\mathcal{H} \geq a^5$; on the second $\mathcal{H} \geq t^4 \geq a^4$. Apply Markov's inequality to (26). For the last bound use $1 - \min(v,b) = \max(|v-1|, |b-1|)$. □

The first two inequalities permit concentration near either endpoint O or C, not just the product pair. Additional normalization, gap control, or geometric compactness would be needed to exclude an approximate O component quantitatively. These conclusions concern Weyl-norm distributions and derivative energies, not Gromov–Hausdorff distance, topology, or arbitrary approximately Einstein metrics. The improvement is a certificate-derived simplification of a specific terminal step, together with explicit error-sensitive consequences.

8.1 An explicit moment-balance error

Proposition 8.4 (Cost of unequal second moments)**.** *Assume* $(v,b) \in \mathcal{D}$ *and the weighted variance inequality* (5)*, but do not assume balanced second moments. Put* $d = v - b$*,* $\varepsilon_2 = \mathbf{E}(v^2 - b^2)$*, and*

$x_0 = \mathbf{E}(\delta d)$. *Then the source variance addition has lower bound*

$$D\mathbf{E}\left[\frac{|\nabla d|^2}{K} - (1-\delta^2/4)d^2\right] \geq \frac{D}{4}\{\mathrm{Var}(\delta d) - 2\varepsilon_2 x_0 - \varepsilon_2^2\}$$
$$\geq -\frac{D}{4}(2|\varepsilon_2| + \varepsilon_2^2).$$

Thus this particular failure of exact moment balance can be charged as an explicit integral error, provided all other identities in the combined integrand remain valid.

Proof. Since $(v+b)d = (2-\delta)d$, one has $2\mathbf{E}d = x_0 + \varepsilon_2$. Subtracting the potential part of the variance addition from (5) gives

$$D\left[\frac{1}{4}\mathbf{E}[(\delta d)^2] - (\mathbf{E}d)^2\right] = \frac{D}{4}\{\mathrm{Var}(\delta d) - 2\varepsilon_2 x_0 - \varepsilon_2^2\}.$$

For $s = v + b \in [0,2]$, $|d| \leq s$, so $|\delta d| \leq (2-s)s \leq 1$. Hence $|x_0| \leq 1$, proving the lower bound. □

Here $|\nabla d| = |p - q|$. This proposition does not assert that the other Einstein tensor identities survive a perturbation of the metric.

9 Equality geometry and non-transfer boundaries

For clarity, the geometric implication after constant norms can be proved without the certificate machinery.

Proposition 9.1 (Conditional splitting after exact norm equality)**.** *Suppose a complete simply connected oriented Einstein four-manifold with* $\mathrm{Ric} = 3g$ *has* $v = b = 1$ *and satisfies the two norm identities* (4) *with* $\sigma, \bar{\sigma} \leq 1$*. Then* T, U *are parallel with spectra* $(2, -1, -1)$*, and the manifold is isometric to* $S^2(1/\sqrt{3}) \times S^2(1/\sqrt{3})$*.*

Proof. The identity for the constant v becomes $0 = 6(1-\sigma) + e$. Both terms are nonnegative, so $e = 0$ and $\sigma = 1$ almost everywhere, hence everywhere by the smooth tensor equation. The exchanged statement gives $\nabla U = 0$. For either block, trace, squared norm, and cubic trace are $0, 6, 6$. Newton's identities give the characteristic polynomial $\lambda^3 - 3\lambda - 2 = (\lambda - 2)(\lambda + 1)^2$.

The simple eigenlines are parallel real line subbundles of Λ^+ and Λ^-. On a simply connected manifold a real line bundle has no sign monodromy, so choose global unit sections $\omega_\pm$. Their derivatives belong to their respective lines and are perpendicular to the unit section, hence vanish. The associated endomorphisms $I = \sqrt{2}\omega_+$ and $J = \sqrt{2}\omega_-$ are parallel orthogonal complex structures of opposite orientations; they commute. Therefore $A_0 = IJ$ is a parallel symmetric involution. Its trace is zero by orthogonality of the two form types, so its two eigenspaces have rank two. They give an orthogonal parallel splitting. The classical de Rham theorem for complete simply connected manifolds yields a global product of complete simply connected surfaces. Each factor has Ricci tensor $3g$, hence constant Gaussian curvature 3, and is the round sphere of radius $1/\sqrt{3}$. □

Three boundaries must remain explicit. First, a zero sectional plane means $x + y = 2$ at one point; it does *not* mean $v + b = 2$ there, let alone everywhere. Second, a strictly positive Einstein theorem cannot be extended by replacing a strict inequality with a weak one: the source needs its own zero-set regularization, semidefinite Hessian square, and equality analysis. Third, the proposition concerns the universal cover and does not classify all finite isometric quotient actions on the product. No quotient enumeration is attempted.

10 Public literature comparison

Public primary sources were consulted by mathematical topic and public title only. The retrieval snapshot is October 8, 2026 UTC. The comparisons below distinguish hypotheses rather than using a similar conclusion as a substitute for a missing implication.

Di Cerbo's revised paper [2], Section 3, Proposition 3.2, gives product rigidity under nonnegative sectional curvature and saturation of the Weyl-norm bound at *every point*. In the present normalization that saturation is $v + b = 2$ everywhere. It does not replace the step from one zero plane to constant norms. Its proof has additional global complex-geometric ingredients, whereas Proposition 9.1 above starts from the stronger conclusion that both Weyl blocks are parallel.

Cheng's September 2026 preprint [3] treats positive sectional curvature under $\chi \leq 3$. Gursky–Malchiodi's September 2026 preprint [4] treats a positive-sectional-curvature result with the topological bound $2\chi - 3|\tau| \leq 4$. Liu's toric theorem [5] assumes a T^2-invariant Einstein metric on a closed simply connected four-manifold. These are useful current comparisons, but none supplies the one-zero-plane implication without its extra hypotheses. The source's strictly positive companion is not adopted here as a black-box boundary theorem.

Ozuch's October 6, 2026 preprint [6] announces positive-scalar-curvature Einstein metrics on $\#_k(S^2 \times S^2)$ and on $k\mathbb{CP}^2\#(2k+1)\overline{\mathbb{CP}}^2$ for $k \geq 2$. This is a newly submitted preprint statement, not a peer-review certification or a proof checked here. It concerns positive Einstein constant, not nonnegative sectional curvature. It is therefore not a contradiction of M669 and is a current warning against dropping the sectional hypothesis in favor of Ricci or scalar positivity.

Classical Bernstein certificates and subdivision were already developed in public literature, including Boudaoud–Caruso–Roy and Roy [7, 8]. The matrix partition-of-unity implication follows by applying scalar positivity to every vector quadratic form. The exact kernel and weighted-margin consequences in Sections 4–6 are local derivations for this report. They do not establish priority over all prior work. The private Ram representation adds explicit domains, normalization, reconstruction identities, zero patterns, and residual budgets to the reporting of these known certificate mechanisms.

11 Verification scope, limits, and further research

The local source inventory contains the complete listed mathematical TEX files and verification documentation. The documentation advertises `verification/verify.py`, `polynomial.py`, and a reference `results.json`, but these executable/data files are absent from the inspected local M669 directory. Their advertised success counts are not an executed result of this report. Exact finite replay must reconstruct the specified arrays from the mathematical inputs or obtain the pinned files independently; absence from this local extraction does not prove absence from the upstream release.

The independent exact reconstruction passed forty certificate groups: eight quartic scalar interval groups, nine matrix-triangle groups, three scalar-triangle groups, and twenty potential-quotient interval groups. It checked coefficient signs, determinant bounds, zero exceptions, the potential expansion and factor identities, and the new vanishing-weight margins. The separate bounded local check used standard-library rational arithmetic to recheck all 349 nonzero $\mathcal{B}$ coefficient matrices after subtraction of the new quadratic weight, all scalar boundary coefficients after subtraction of the fourth-order weight, the five common-potential coefficient dominations, and the active-weight inequalities. These checks use exact rational coefficient arrays, not a grid.

A separate bounded exact replay recorded 120 passing checks across M669 and its strict-curvature companion. Its M669-relevant portion verifies all three upper-volume polynomial integral bounds and their total; all 127 coefficients of the seven constrained-moment Bernstein certificates (degrees

8, four times 17, and twice 22); and the recorded derivative, endpoint, and clearing-denominator identities. The two lower-volume integral rows in that replay belong only to the companion, not to M669. This is attributed execution evidence with captured script, results, read scope, and source hashes; it is not a second execution by this report. The analytic sine comparison, rational brackets for π, measure-separation argument, and geometric comparison are not certified by those finite operations.

The local results have distinct verification levels. Theorem 4.1, Theorem 7.1, and Proposition 9.1 have complete proofs under their stated hypotheses. The finite hypotheses of Theorems 5.1 and 6.1 have been independently replayed. Theorems 5.2 and 8.2 still require the source's analytic and tensor-to-polynomial reductions. Neither replay verifies the global volume comparison, representation-theoretic Hessian identities, or complete source theorem. A separate displayed-derivation review found no corrective theorem-level issue in the transfer and terminal argument, but was not a formal or exhaustive verification of those upstream inputs. Exact scope and hashes are retained in the accompanying audit.

The benefits are specific: active coefficient sets preserve the boundary geometry of a certificate; exact determinant-to-trace margins avoid unreliable floating-point eigenvalues; the common potential replaces the strict level-set argument at the terminal step; and the resulting deficit inequality quantifies spectral concentration and gradient energy where the source gave a zero-defect conclusion. The costs are also specific: the shorter terminal proof retains all upstream analytic assumptions, boundary zeros demand exact reconstruction or structured errors, several finite checks can still involve large rational numerators, and a norm residual does not control the metric. No running-time gain, optimal coefficient choice, or automatic discovery guarantee is asserted.

The next mathematical tasks are to connect independently replayed coefficient margins to an end-to-end residual bound for the original weighted integral identity; to justify any proposed perturbation class for actual metrics; and to determine whether additional elliptic or noncollapsing information can upgrade the weighted fourth-power bound to geometric closeness. Each requires new work. A small numerical error alone cannot close an equality case, and neither a better numerical factor nor a longer certificate establishes a new global theorem.

A Finite check specifications

A bounded checker for the local extensions should perform the following exact tasks, and no search over arbitrary metrics is implicit.

1. Reconstruct (13), E_0, c, Θ and compare $cE_0 - \Theta^2$ after substitution with (18), coefficient by coefficient.

2. Form h_i, divide by the explicit d_i with zero polynomial remainder, and check (19) by the one-dimensional version of (10) on each quarter interval.

3. For the matrix consequence, verify the exact zero indices and all nonzero PSD margins in the six prescribed arrays, then compute a rational minimum of $\det C / \operatorname{tr} C$. This is verification of supplied data, not coefficient optimization.

4. Check the binomial identities used in the two weight bounds, the exact tangential potential polynomial, and the rational-function identity after substituting $v = u^2$.

5. Keep every rational identity, reported bound, exceptional index, and affine domain in the check record, including failed or missing assertions. A finite sampling grid is not a proof of nonnegativity.

The main certificate work is logically separable from checking these local identities. The accompanying audit specifies which checks were actually executed, their output, and their hashes. No check listed only as a specification is represented as completed.

References

[1] OpenAI, *Zero-Plane Rigidity for Einstein Four-Manifolds*, October 4, 2026, pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, source directory identified in Section 1.

[2] L. F. Di Cerbo, *A rigidity theorem for Einstein 4-manifolds with sectional curvature of a fixed sign, and its consequences*, arXiv:2503.09570v3, July 22, 2026; Math. Ann. 396 (2026), article 11. https://arxiv.org/html/2503.09570v3.

[3] L. Cheng, *On Positively Curved Einstein Four-Manifolds with Euler Characteristic at Most Three*, arXiv:2609.23337v1, September 20, 2026. https://arxiv.org/abs/2609.23337v1.

[4] M. J. Gursky and A. Malchiodi, *Einstein four-manifolds of positive sectional curvature and ADM mass*, arXiv:2609.30803v1, September 25, 2026. https://arxiv.org/abs/2609.30803v1.

[5] T. Liu, *Toric Einstein 4-manifolds with nonnegative sectional curvature*, arXiv:2311.04840; Ann. Global Anal. Geom. 67 (2025), article 14. https://arxiv.org/abs/2311.04840.

[6] T. Ozuch, *Positive Einstein metrics on infinitely many four-manifolds*, arXiv:2610.08052v1, submitted October 6, 2026, new preprint. https://arxiv.org/abs/2610.08052.

[7] F. Boudaoud, F. Caruso, and M.-F. Roy, *Certificates of Positivity in the Bernstein Basis*, Discrete Comput. Geom. 39 (2008), 639–655. https://doi.org/10.1007/s00454-007-9042-x.

[8] M.-F. Roy, *Certificates of Positivity in the Bernstein basis*, Oberwolfach Reports 14/2007, 800–804, joint work with F. Boudaoud, F. Caruso, and R. Leroy. https://ems.press/content/serial-article-files/46100.

[9] M. J. Gursky and C. LeBrun, *On Einstein manifolds of positive sectional curvature*, Ann. Global Anal. Geom. 17 (1999), 315–328. https://doi.org/10.1023/A:1006597912184.

[10] M. Subbarao, private Ram Calculus corpus, Volume 16, *Polynomial enclosures in Bernstein form*; exact local path and labels in Section 3.

[11] M. Subbarao, private report `TR-item-047-astra.tex`, *Global Optimality Sandwich* and *Residual-Aware SOS Lower Bounds*; exact local path and sections in Section 3.

[12] M. Subbarao, private report `TR-item-305-astra.tex`, rational facial certificates, repair, weak infeasibility, and irrational exposure; exact local path and sections in Section 3.